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Circulating cell free DNA (ccfDNA) fragmentomics for early detection of colorectal carcinoma: A systematic review and meta-analysis.

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Background: The high morbidity and mortality associated with colorectal carcinoma (CRC) is linked primarily to its late detection. Current screening methods for CRC have limitations with regards to its invasiveness. Liquid biopsy, particularly cell-free DNA (ccfDNA) fragmentomics, has emerged as a promising, non-invasive approach for early CRC detection and stratification, capable of distinguishing CRC patients from healthy individuals with high sensitivity and specificity. ccfDNA fragmentomics has been shown to outperform serum protein biomarkers in the detection of CRC. This approach has been shown to be particularly effective with small-sized, early-stage tumours. It has also been found that the accuracy of detection of CRC is improved when ccfDNA fragmentomics is used in conjunction with other biomarkers such as protein, genetic features of ccfDNA, and methylation signatures. These advances suggest that ccfDNA fragmentomics, especially when combined with other molecular features, holds significant promise for non-invasiveness and early detection of CRC. Therefore a systematic review is necessary to consolidate the existing evidence and determine the most effective approach to utilise ccfDNA fragmentomics in the early detection of CRC. **Methods:** A systematic search was conducted using PubMed, Google Scholar, and Scopus. PRISMA guidelines were followed. A boolean expression was constructed to retrieve and select articles from major medical databases. Articles that involved (ccf-mtDNA) assay along with measures of diagnostic accuracy i.e. Sensitivity and Specificity were included in the study. The R Studio package was used to evaluate the potential of the diagnostic test. The Pooled sensitivity and specificity were estimated to predict the diagnostic potential of ccf-mtDNA in the early detection of colorectal cancer. The random effects model via the linear (mixed-effects) model framework was considered for statistical analysis. **Results:** A total of 5 studies with 2553 observations where detection of ccf-mtDNA was considered to diagnose colorectal cancer were assessed through this meta-analysis. The Pooled Sensitivity and Specificity were estimated to be 94.50% ([92.35; 97.51], 95% CI, $p < 0.0001$, $z = 76.6418$, $SE = 0.1225$) and 89.15% ([77.74; 98.7], 95% CI, $p < 0.0001$, $z = 15.2693$, $SE = 0.06$) utilising the random effects model. The pooled Area under the curve was estimated to be 0.9864(0.9818;0.9909, 95% CI, $p < 0.0001$, $z = 424.7193$, $SE = 0.0023$), indicating high accuracy and consistency. **Conclusions:** The ccf-mtDNA, owing to its higher sensitivity, specificity and accuracy can be a potential biomarker in early diagnosis of colorectal cancer., highlighted by this meta-analysis recent advances, boost the diagnostic value of the ccf-mt DNA paving the road to a highly compliant and accurate non-invasive detection method for early diagnosis of Colorectal cancer. Research Sponsor: None.

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